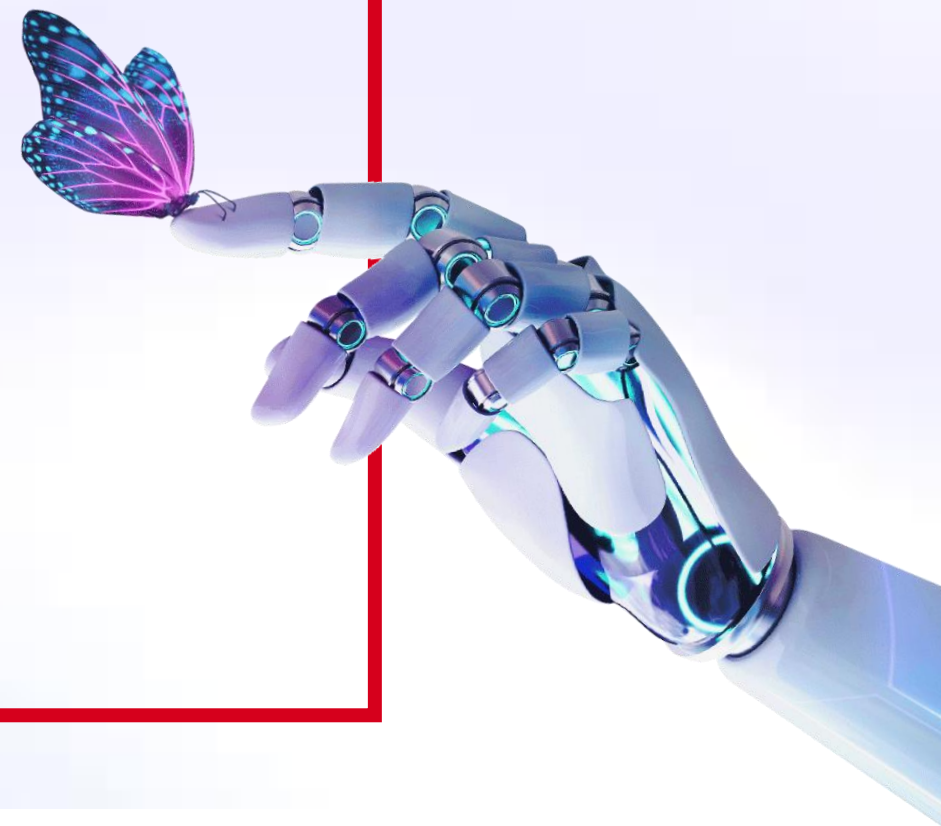


Evaluating Azure Container Service options



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May 2025

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CONTAINERS

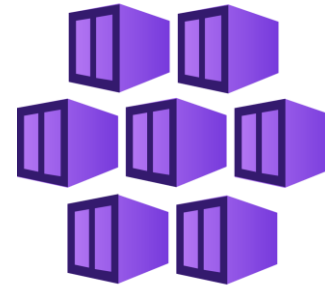
Options to host containers in Azure



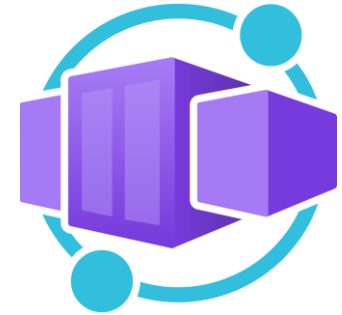
Web App (for Container)



Container Instances

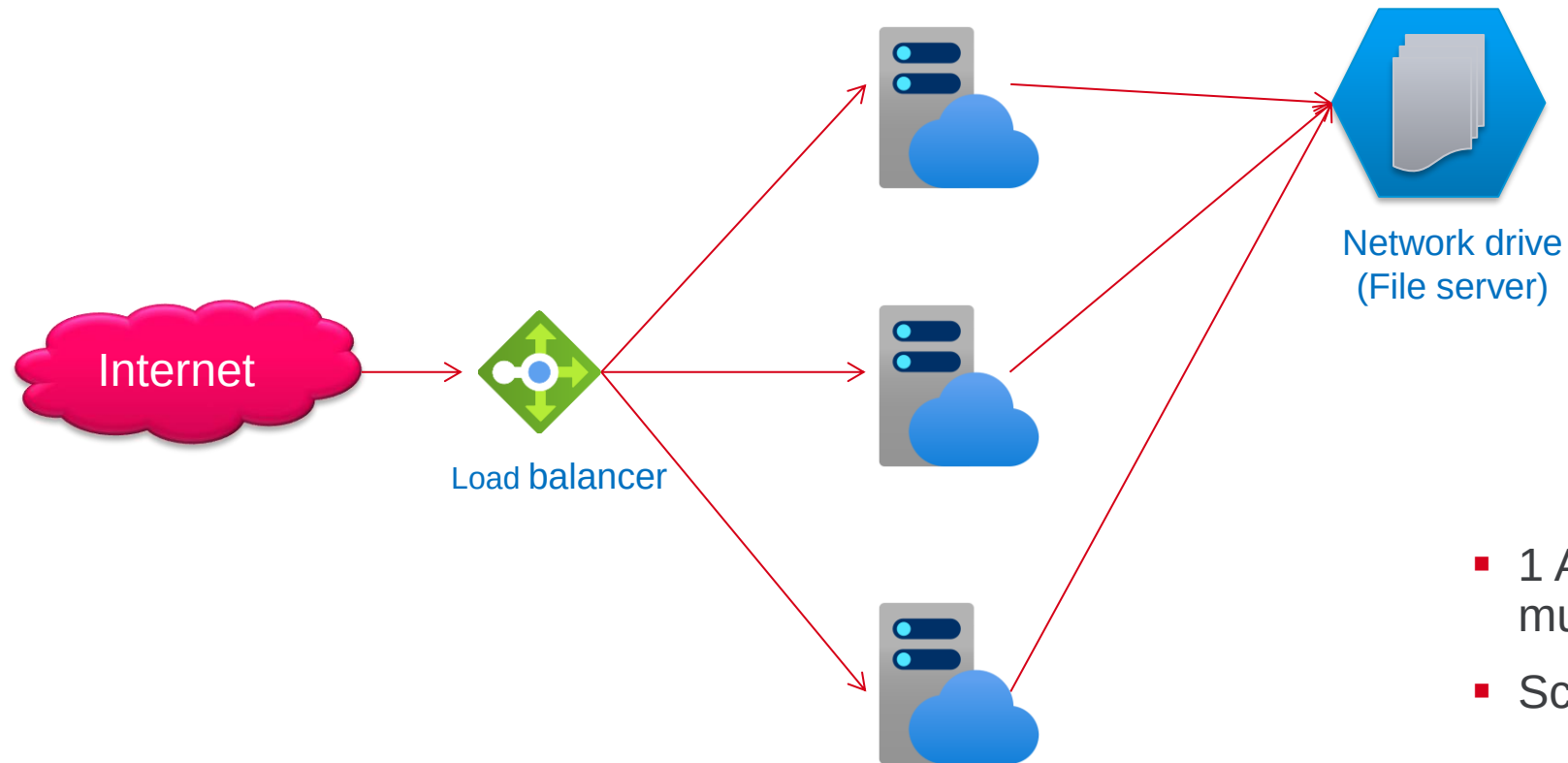


Kubernetes Service



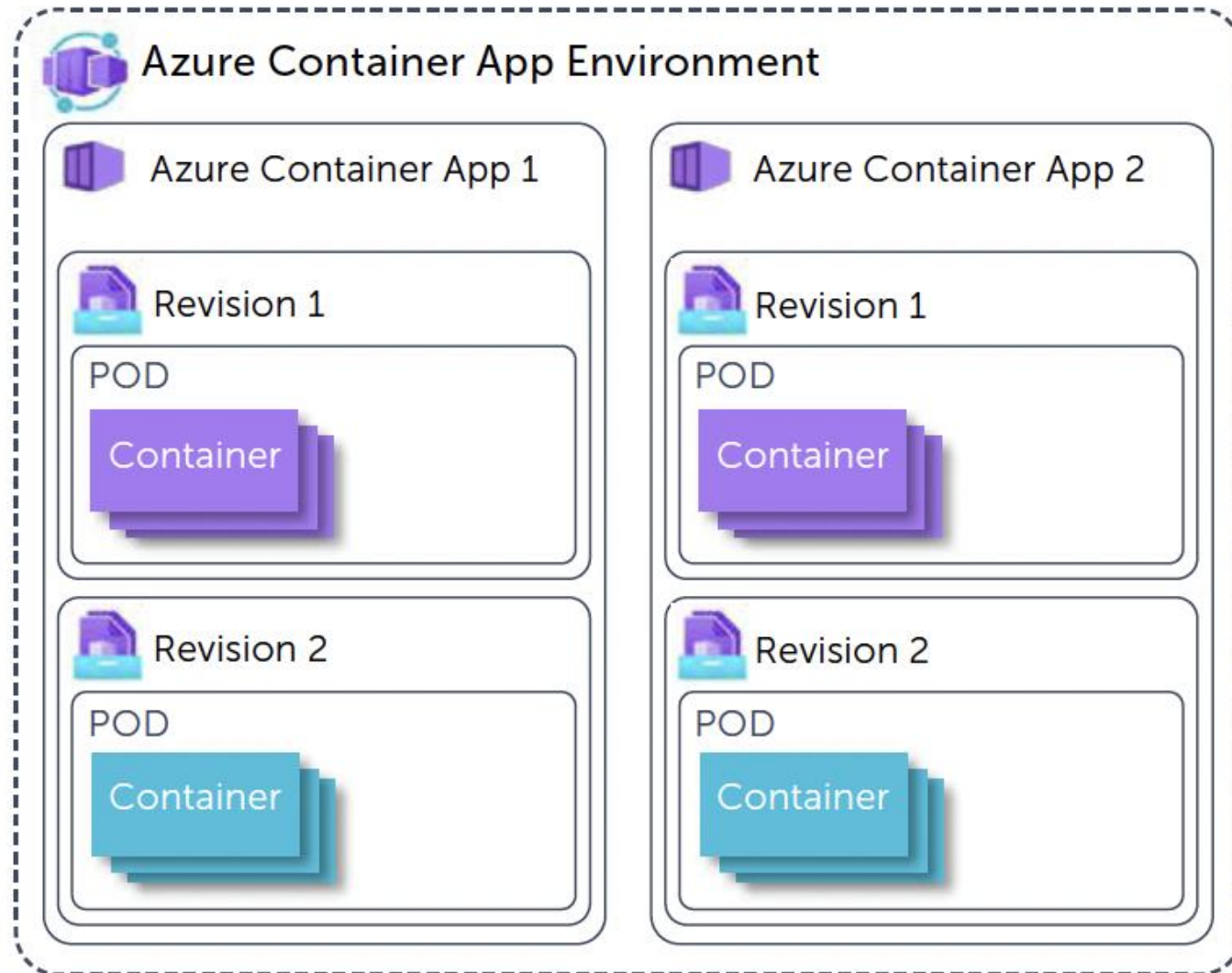
Container Apps

Web App Architecture



- 1 App Service Plan can have multiple web apps
- Scale unit is VM

Container Apps

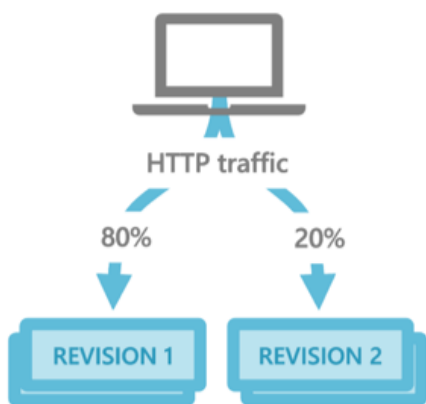


- Based on Kubernetes
- Scale unit: number of pod (replica)
- Consumption plan (serverless): you pay for numbers of pod (replicas)
- Dedicated plans: pay per number of VM nodes



Azure Container Apps: Example scenarios

PUBLIC API ENDPOINTS



HTTP requests are split between two versions of the container app where the first revision gets 80% of the traffic, while a new revision receives the remaining 20%.

BACKGROUND PROCESSING



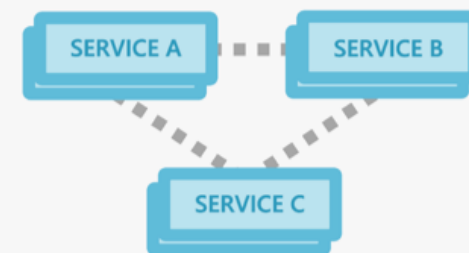
A background job that transforms data in a database.

EVENT-DRIVEN PROCESSING



A queue reader application that processes messages as they arrive in a queue.

MICROSERVICES



Deploy and manage a microservices architecture with the option to integrate with Dapr.

AUTO-SCALE CRITERIA

Scaling is determined by the number of concurrent HTTP requests.

AUTO-SCALE CRITERIA

Triggered by a schedule, on demand, or based on events.

AUTO-SCALE CRITERIA

Scaling is determined by the number of messages in the queue.

AUTO-SCALE CRITERIA

Individual microservices can scale according to any KEDA scale triggers.

Web App vs Container Apps

Feature	Web App	Container Apps
Foundation	Virtual Machine	Kubernetes
Scaling	Via Service plan, all apps the service plan will be scaled together	By replica, independency for each app. Support scale to zero KEDA
Resource limit (CPU/RAM) for each app	No	Yes
OS Support	Windows, Linux	Linux Only
Private Apps	Yes, via Private Endpoint	Yes, natively without Private Endpoint
Support microservices	Limited	Good, Dapr

AKS vs ACA

Feature	AKS	ACA
Simplicity	Required Kubernetes knowledge	Very simple
Cluster management	Partially (you manage more)	Fully managed by Azure
Scaling	Manual or auto-scaling, but requires configuration	Built-in automatic scaling (including scale to zero)
Team skill required	High	Low
Level of customization	High	Low

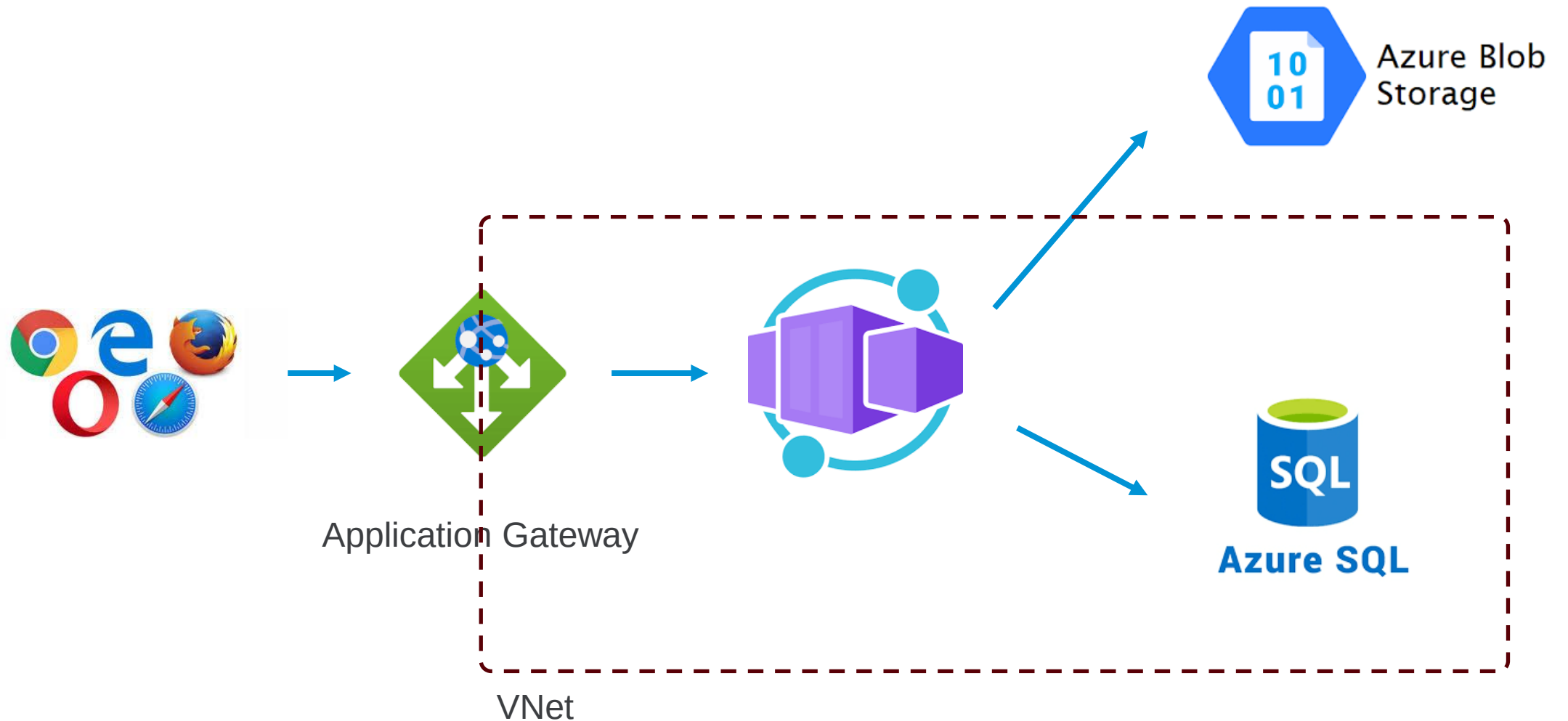
How about the cost

container comparision

					   
✓ Virtual Machines		1 D4a v4 (4 vCPUs, 16 GB RAM) x 730 Hours (Pay as...	 	Upfront: \$0.00	Monthly: \$140.16
✓ App Service		Standard Tier; 1 S3 (4 Core(s), 7 GB RAM, 50 GB Stor...	 	Upfront: \$0.00	Monthly: \$277.40
✓ Azure Kubernetes Service (AKS)		Standard; Cluster management for 1 clusters; 1 D4a ...	 	Upfront: \$0.00	Monthly: \$213.46
✓ Azure Container Apps		Dedicated Plan Type, Pay as you go, 1 X D4, 4 vCPU,...	 	Upfront: \$0.00	Monthly: \$224.81

Link to Azure Pricing Calculator <https://azure.com/e/20a4c3bee2854dfe8801644e7234f126>

SimplCommere Demo



Thank you